

HW 4.3 Reproducible Data Analysis

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Busiest Airports Analysis

Table

Table 1: Table 1: Passenger counts in millions for six major airports from 2020 to 2024.

Airport	IATA	Year	Total Passengers (Millions)
Hartsfield-Jackson Atlanta Intl	ATL	2020	42.919
Hartsfield-Jackson Atlanta Intl	ATL	2021	75.705
Hartsfield-Jackson Atlanta Intl	ATL	2022	93.700
Hartsfield-Jackson Atlanta Intl	ATL	2023	104.653
Hartsfield-Jackson Atlanta Intl	ATL	2024	108.068
Denver International Airport	DEN	2020	33.741
Denver International Airport	DEN	2021	58.829
Denver International Airport	DEN	2022	69.286
Denver International Airport	DEN	2023	77.838
Denver International Airport	DEN	2024	82.359
Dubai International Airport	DXB	2020	25.837
Dubai International Airport	DXB	2021	29.111
Dubai International Airport	DXB	2022	66.070
Dubai International Airport	DXB	2023	86.994
Dubai International Airport	DXB	2024	92.332
Frankfurt Airport	FRA	2020	18.769
Frankfurt Airport	FRA	2021	24.813
Frankfurt Airport	FRA	2022	48.918
Frankfurt Airport	FRA	2023	59.355
Frankfurt Airport	FRA	2024	61.565
Heathrow Airport	LHR	2020	22.111
Heathrow Airport	LHR	2021	NA

Airport	IATA	Year	Total Passengers (Millions)
Heathrow Airport	LHR	2022	61.615
Heathrow Airport	LHR	2023	79.183
Heathrow Airport	LHR	2024	83.885
Beijing Daxing Intl	PKX	2020	NA
Beijing Daxing Intl	PKX	2021	25.051
Beijing Daxing Intl	PKX	2022	NA
Beijing Daxing Intl	PKX	2023	NA
Beijing Daxing Intl	PKX	2024	49.441

The table summarizes passenger traffic for six major airports between 2020 and 2024. Atlanta remains one of the busiest airports throughout the period, while Dubai and Heathrow also show strong passenger totals in the later years. Some airports have missing values in certain years, which should be considered when comparing trends across the full period.

Plot

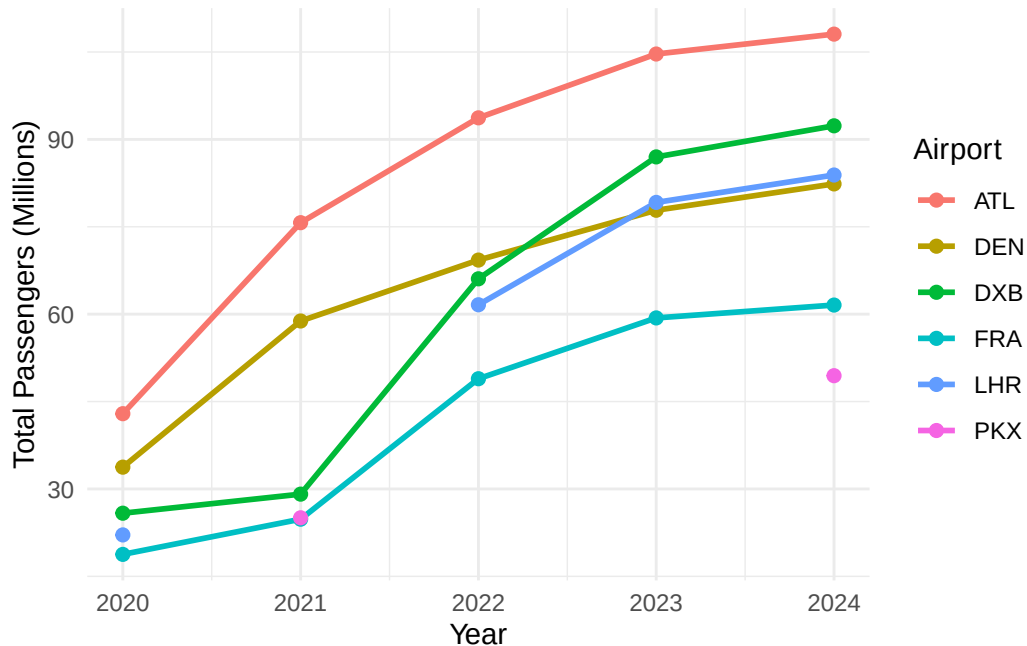


Figure 1: Figure 1: Passenger traffic trends for six major airports from 2020 to 2024.

The plot makes the overall trends easier to compare across airports. Most airports show strong growth after 2020, which suggests recovery in air travel following the pandemic period.

Atlanta, Dubai, and Heathrow reach especially high passenger levels by 2023 and 2024, while Beijing Daxing appears less consistent because data are missing for multiple years. Compared with the table, the plot makes the pace of recovery much easier to see.

Monte Carlo Numerical Integration

Small Multiple Plot

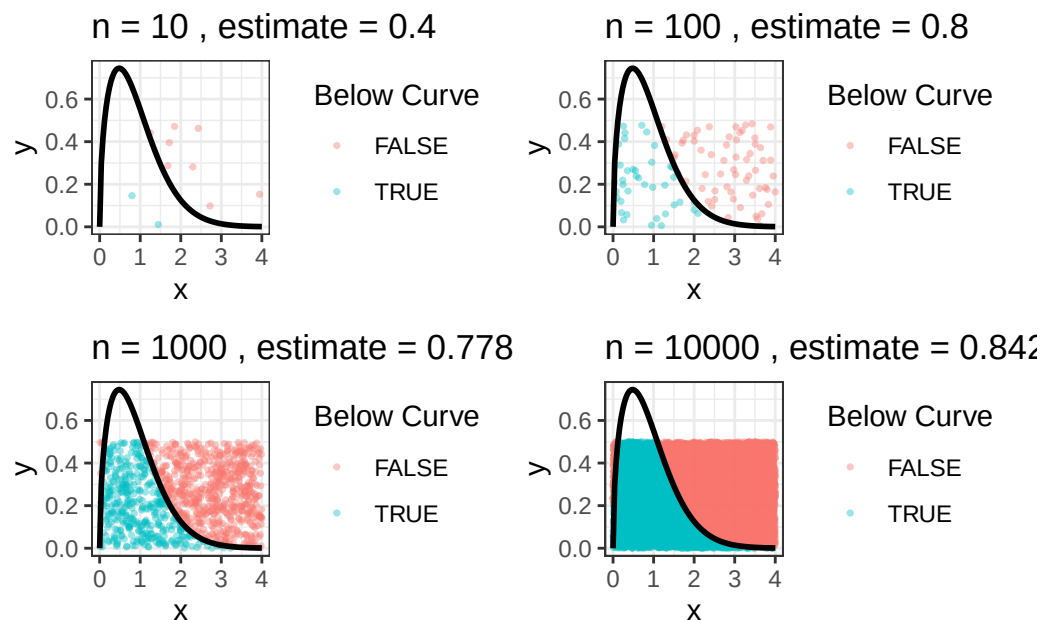


Figure 2: Monte Carlo simulation plots for four different sample sizes.

Monte Carlo numerical integration estimates the area under a curve by randomly generating points inside a rectangle and checking what proportion of them fall below the function. In this case, the function is the Weibull density with x-values from 0 to 4 and y-values from 0 to 0.5. The area of the rectangle is therefore 2, so the proportion of points below the curve is multiplied by 2 to estimate the integral.

The small multiple plot shows how the estimate changes as the number of simulated points increases. With only 10 points, the estimate is rough and highly variable, but as the sample size increases to 100, 1000, and 10000, the point pattern follows the curve more clearly and the estimate becomes more stable. This supports the idea that Monte Carlo integration becomes more reliable as the number of random points increases.

Generative AI Reflection

Use of Generative AI

Generative AI tools such as ChatGPT were used mainly to verify the correctness of my work and check for any missing components. The code and analysis were written independently, and AI was only used to review structure, catch small errors, and confirm that the requirements of the assignment were met.

Evaluation of AI Output

The AI outputs were useful for identifying gaps or minor mistakes, but they were not always directly applicable. Some suggestions needed to be adjusted to match the level and expectations of the course. This required reviewing and refining the information before using it.

Reflection

Using generative AI as a verification tool helped ensure that my work was complete and correctly structured. It was most helpful for catching small issues and confirming that I was on the right track. However, the main understanding and implementation came from working through the assignment independently.